WORKED SOLUTION

STEP I 2000, Question 13

Probability and Statistics • Conditional Probability

STEP I 2000, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q13

The key idea

Everything comes down to tracking one gene from each parent. In (i) each child of two AB parents is AA or BB with the same small probability, and the four children are independent, so we count arrangements and multiply. Part (ii) is a "reverse" conditional probability: we know the outcome (the child) and want the cause (the father). That is Bayes' theorem, and a tree diagram over the father's three possible genotypes organises it.

Solution

Part (i)

Each parent is AB, so each passes on A or B with probability $\frac{1}{2}$, and the two parents act independently. For a single child,

$$\mathrm{P}(AA) = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}, \qquad \mathrm{P}(BB) = \tfrac{1}{4}, \qquad \mathrm{P}(AB) = \tfrac{1}{2},$$

where the last one counts both orders (A from the mother and B from the father, or the other way round).

Different children inherit independently of one another. A particular ordering of the four children with two AA and two BB, such as AA, AA, BB, BB, has probability $\left(\frac{1}{4}\right)^4$. The number of ways to choose which two of the four children are the AA ones is $\binom{4}{2} = 6$. So

$$\mathrm{P}(\text{two } AA \text{ and two } BB) = 6 \times \left(\tfrac{1}{4}\right)^4 = \frac{6}{256} = \frac{3}{128},$$

as required.

Part (ii)

Let F_{AA}, F_{AB}, F_{BB} be the events that my father is AA, AB, BB, and let C be the event that I am AA. We take my father to be a random member of the population, chosen

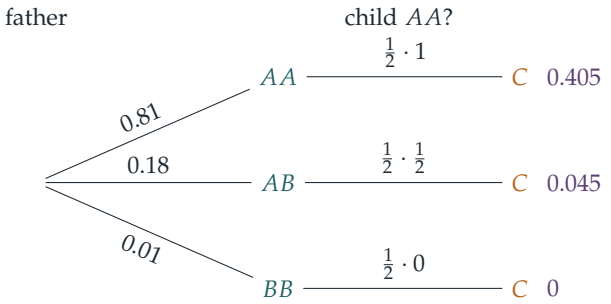
independently of my mother's genotype, so

$$\mathrm{P}(F_{AA}) = 0.81, \qquad \mathrm{P}(F_{AB}) = 0.18, \qquad \mathrm{P}(F_{BB}) = 0.01.$$

Ask yourself Why is 0.18 not already the answer?

Because we have extra information: I am AA. That rules out a BB father completely and makes an AA father more likely, since an AA father always passes on A. So we must condition on C.

For me to be AA I need an A from my mother and an A from my father. My mother is AB, so she gives A with probability $\frac{1}{2}$ whoever my father is. The father gives A with probability 1, $\frac{1}{2}$ or 0 according to his genotype. The tree:



By the law of total probability,

$$\begin{aligned}\mathrm{P}(C) &= 0.81 \cdot \tfrac{1}{2} \cdot 1 + 0.18 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} + 0.01 \cdot \tfrac{1}{2} \cdot 0 \\ &= 0.405 + 0.045 + 0 = 0.45.\end{aligned}$$

Then by Bayes' theorem,

$$\mathrm{P}(F_{AB} \mid C) = \frac{\mathrm{P}(F_{AB})\,\mathrm{P}(C \mid F_{AB})}{\mathrm{P}(C)} = \frac{0.045}{0.45} = \frac{1}{10}.$$

So the probability that my father is AB is $\frac{1}{10}$.

Checks

- In (i), the probabilities for the four children's genotype counts should total 1; the $\binom{4}{2}$ factor is the same one that appears in the multinomial expansion of $\left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right)^4$, of which $3/128 \approx 0.0234$ is one term. An exact enumeration of all 2^8 ways the eight genes can be inherited gives $3/128$.
- In (ii), the answer has dropped from the prior 0.18 to 0.1, as it should once a BB father is excluded and an AA father is favoured. A simulation of 10^6 families gave 0.0998.

Common slips

- Forgetting the factor $\binom{4}{2} = 6$ in (i), and getting $1/256$.
- Answering 0.18 in (ii), or computing $\mathrm{P}(C \mid F_{AB}) = \frac{1}{4}$ instead of $\mathrm{P}(F_{AB} \mid C)$.
- Leaving out the mother's factor $\frac{1}{2}$ in only some branches: it cancels in the end, but only if it is used consistently.

Beyond STEP

The population proportions 81%, 18%, 1% are p^2, $2pq$, q^2 with $p = 0.9$, $q = 0.1$: the Hardy–Weinberg equilibrium, which is preserved from generation to generation under random mating. Inferring a hidden cause from an observed outcome is the heart of Bayesian statistics, and the same calculation underlies genetic counselling and paternity testing. Blitzstein and Hwang, *Introduction to Probability*, has a thorough treatment of conditioning and Bayes' theorem with many examples like this one.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.